

TASK 2 — Virtual Try-On AI Models

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Date: 14/05/2026

Project Folder: Raritone

1. Introduction

Virtual try-on systems are advanced AI-based applications that allow users to digitally try clothes without physically wearing them. These systems are widely used in e-commerce to improve user experience.

2. Working Mechanism

2.1 Pose Detection

The system first detects the human body and identifies key landmarks such as shoulders, arms, and torso.

2.2 Body Segmentation

The human body is separated from the background using segmentation techniques.

2.3 Clothing Overlay

Clothing items are digitally overlaid onto the detected body structure.

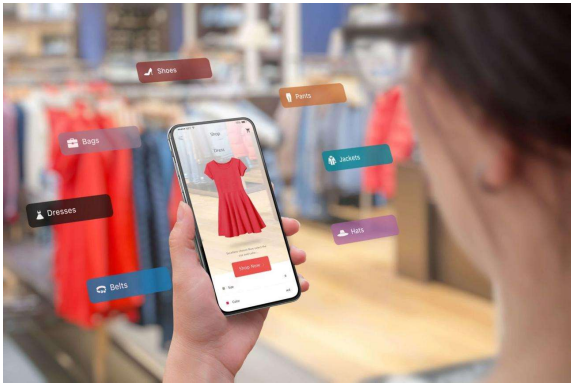
2.4 Alignment and Scaling

The clothing is adjusted according to body size and orientation.

3. Technologies Used

- Computer Vision
- Deep Learning
- Augmented Reality
- Pose Detection frameworks

4. Visualization



5. Advantages

- Enhances online shopping experience
- Reduces return rates
- Saves time
- Provides realistic preview

6. Challenges

- Accurate body measurement
- Realistic cloth simulation
- Handling different poses and lighting

7. Conclusion

Virtual try-on systems represent a significant advancement in AI applications for fashion, providing convenience and improving customer satisfaction.